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IRIS

PROJECT PROPOSAL

IRIS: Integrated Reporting Intervention System Temporal Graph-Based Surveillance Recovery and Intervention Optimization for Philippine Health Reporting Networks

TEAM NEX

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EXECUTIVE SUMMARY

IRIS (Integrated Reporting Intervention System) is a temporal graph-based decision-support system designed to help Philippine health surveillance units detect and address disruptions in disease reporting. Incomplete, delayed, or inconsistent reports can compromise surveillance data quality and public health decision-making (World Health Organization [WHO], 2017).

Designed for Regional Epidemiology and Surveillance Units (RESUs), IRIS analyzes reporting patterns and network relationships to identify potential shared disruptions, support investigations, and prioritize corrective interventions based on available resources. Temporal graph modeling enables the analysis of interconnected entities and their changing behavior over time (Kazemi et al., 2020).

The system will be evaluated using historical disease datasets and synthetic reporting networks, comparing its detection accuracy, false-alarm rates, and simulated recovery efficiency against conventional methods.

Ultimately, IRIS aims to strengthen surveillance reporting reliability by transforming detected anomalies into actionable, evidence-based interventions.

KEYWORDS

Temporal Graph, Surveillance Data, Reporting Anomalies, Shared Network Disruptions, Intervention Optimization

BACKGROUND

Reliable disease surveillance depends on complete, timely, and consistent reporting. However, unusual case trends may reflect either actual epidemiological changes or reporting inaccuracies caused by missing submissions, delays, or disruptions, potentially affecting public health decisions (WHO, 2017; WHO, 2018).

In the Philippines, Regional Epidemiology and Surveillance Units (RESUs) manage surveillance activities, support outbreak investigations, and provide timely health information (DOH–CALABARZON, n.d.). However, reporting practices and available data differ across diseases, making it challenging to assess reporting reliability and identify potential surveillance gaps consistently (WHO, n.d.-a, n.d.-b).

Temporal graph modeling enables the analysis of interconnected reporting entities and how their relationships and reporting patterns evolve over time (Zheng et al., 2025). IRIS proposes applying this approach to detect potential shared reporting disruptions, distinguish isolated issues from network-wide problems, and evaluate whether network-aware intervention prioritization can improve surveillance recovery efficiency compared with investigating facilities individually.

THE PROBLEM

Low disease counts do not always indicate fewer infections, as missing submissions, reporting delays, and disruptions may produce misleading surveillance data. Investigating affected facilities individually can also waste resources when multiple reporting problems originate from a shared source (WHO, 2017; WHO, n.d.-a).

IRIS addresses this problem through graph-aware analysis to identify potential shared disruptions and prioritize corrective interventions, while ensuring that reporting failures are verified by authorized personnel.

OBJECTIVES

IRIS aims to develop a temporal graph-based system that detects disease reporting failures and prioritizes corrective interventions to improve surveillance reliability. Specifically, the study aims:

1. To develop a temporal graph model to detect reporting anomalies and shared network disruptions across multiple diseases.
2. To design an intervention optimization mechanism to recommend corrective actions based on identified issues and available resources.
3. To evaluate the system's accuracy and effectiveness in detecting reporting failures and improving simulated reporting recovery.

SCOPE & LIMITATIONS

Primary intended users are regional epidemiologists and surveillance analysts in Philippine RESUs. The prototype supports multi-disease historical analysis, graph-based investigation, verification planning. Public WHO case and notification datasets will be used only at their published geographic and temporal resolution; facility-level network relationships, submission logs and intervention outcomes will be synthetic unless suitable authorized data become available.

The system will not diagnose disease, infer unobserved case totals as fact, certify underreporting, automatically direct medical interventions or claim real-world recovery effects. Geographic adjacency must not be presented as a verified reporting dependency. Missing values must remain distinct from confirmed zero-case reports. Comparisons with baselines and uncertainty disclosures are required before interpreting graph-derived flags.

METHODOLOGY

Data Acquisition and Harmonization

Retrieve historical measles and tuberculosis data from WHO, supplemented by Philippine disease datasets if they provide sufficient temporal coverage, compatible geographic information, and consistent reporting indicators. Standardize disease identifiers, reporting periods, and geographic units while assessing missing values, duplicates, and data completeness. Preserve source metadata and distinguish missing records from confirmed zero-case reports (WHO, n.d.-a, n.d.-b).

Temporal Graph Construction

Construct a synthetic reporting network of health facilities, laboratories, and reporting hubs using $G_t = (V_t, E_t, X_t)$, where nodes represent reporting entities, edges define operational relationships, and features capture reporting behavior over time. Simulate isolated failures, shared-hub disruptions, and recovery events with known outcomes. Keep synthetic network data separate from actual disease records

Anomaly Modeling and Verification

Apply independent time-series models and graph-aware analysis to detect unusual reporting patterns and identify potential shared disruptions. Compare graph-based results with conventional methods using temporal validation and graph ablation. Present anomaly evidence and supporting indicators for verification by authorized surveillance personnel.

Intervention Optimization and Evaluation

Use resource-constrained optimization to prioritize corrective actions based on verified reporting issues, available personnel, intervention costs, and estimated recovery benefits. Compare intervention strategies against conventional facility-level prioritization. Evaluate detection performance using precision, recall, F1-score, and false-alarm rates, alongside shared-source identification and simulated recovery efficiency.

Expected Research Output

Develop an interactive Streamlit or React-based prototype featuring temporal graph visualization, prioritized investigations, evidence-based verification, intervention planning, and reporting recovery assessment. Provide documented datasets, model configurations, evaluation results, and limitations to ensure reproducibility and transparency.

REFERENCES

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APPENDICES

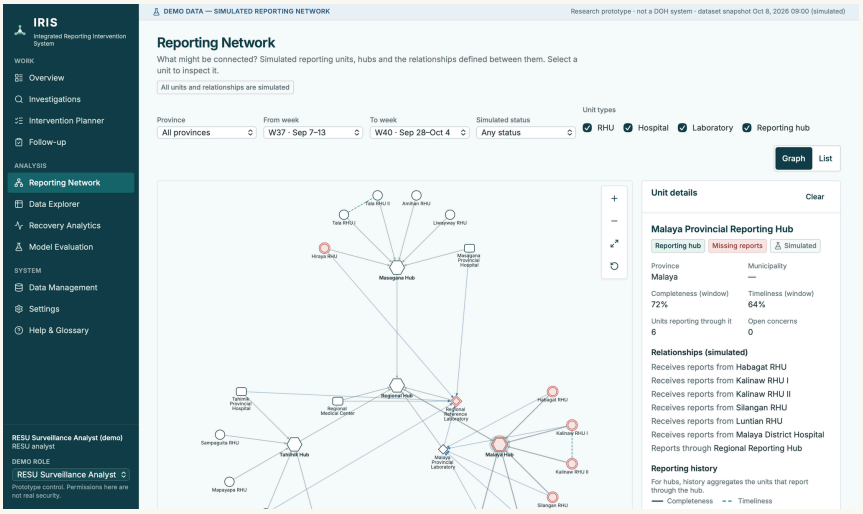
Appendix A. Selected Datasets and Their Uses

Dataset	Intended use in IRIS
Philippine Dengue Cases and Deaths (HDX/DOH)	Analyze regional and provincial dengue trends and identify unusual changes in reported cases and deaths.
COVID-19 DOH Data Drop (Philippines)	Explore historical case distributions, testing facilities, hospital occupancy, and outcomes for additional surveillance analysis.
WHO Measles — Number of Reported Cases	Provide long-term international disease-reporting trends for supplementary analysis.

Appendix B. Evaluation Summary

- **Detection metrics:** Precision, recall, F1-score, and false-alarm rate.
- **Timeliness:** Detection delay.
- **Intervention performance:** Simulated recovery time and recovered submissions per resource unit.
- **Baselines:** Fixed-threshold detection, independent time-series analysis, and facility-first intervention prioritization.
- **Validation:** Chronological testing and graph ablation to assess the contribution of network information.

Appendix C. Proposed System Mockup



Dashboard elements:

- Reporting overview and completeness trends
- Priority investigations and verification status
- Reporting-network visualization
- Intervention planning and recovery monitoring